Morgan Wang

m668wang@uwaterloo.ca | 437-603-2779 | GitHub | LinkedIn | https://portfolio.morgan-wang.workers.dev/

Technical Skills

Design and Analysis: Altium Designer, KiCad, LTspice, PCB Design, Microsoft 365, FDM 3D Printing
Hardware: Arduino UNO, ESP32, Soldering, Circuit Prototyping, Sensor Interfacing, Analog/Digital I/O, I2C, UDP/Wi-Fi
Languages and Tools: C++, Python, SQL, SQLite, Java, Git, GitHub, Arduino IDE, DigiKey

Projects

ESP32 Wireless Telemetry Prototype — October 2026

ESP32, C++, Python, SQL, SQLite

- Built an ESP32 telemetry prototype for the UW Formula Electric TCU redesign, streaming sensor data over Wi-Fi via UDP at ~17.5 packets/s to a Python logger storing every packet in SQLite.
- Ran 18 range tests (~8,300 packets) across 6 distance and obstruction conditions, using sequence numbers and SQL queries to measure packet loss, signal strength and dropout length.
- Found loss stayed ≤ 0.4% at most positions up to 15 m, but reached 12% at one 10 m spot and 42.7% behind a closed door, with dropouts up to 5.8 s, showing the TCU needs a long-range radio and onboard buffering.

5 V DC-DC Buck Converter — September 2026 - October 2026

LTspice, Altium Designer, PCB Design

- Designed a 10–15V to 5V, 1A DC-DC buck converter PCB using an LT1375 switching regulator, a 15 µH inductor, diodes, input/output capacitors, and resistive feedback network through circuit analysis and datasheet calculations.
- Designed and routed a compact 41.4 × 23.6 mm two layer SMD PCB in Altium Designer, incorporating 25 mil power traces, 10 mil signal traces, a bottom layer GND polygon pour, and via connected ground returns.
- Modeled the effect of lower output capacitor ESR (0.10 → 0.05 Ω) and tuned feedback compensation (1 kΩ / 3.3 nF) in LTspice, reducing simulated output ripple from 45.7 mVpp to 23.6 mVpp (48.3%), achieving 32.7% lower ripple than the manufacturer's reference under matched 15V input and 1A load conditions.

Insulation Monitoring Device (IMD) Test Box - UW Formula Electric — September 2026 - October 2026

Altium Designer, Safety Analysis, High Voltage Electronics

- Designed an IMD test box for UW Formula Electric, interfacing with the vehicle's HV+/HV− TSMP and chassis ground in Altium to simulate PASS, WARNING, TECH FAIL, and hard short conditions for FSAE high voltage safety verification.
- Developed the Altium Designer schematic and resistor networks to meet the team's target fault resistances (285 kΩ, 485 kΩ, 75 kΩ), accounting for the 15 kΩ TSMP protection resistance.
- Analyzed high voltage insulation requirements for a 588V tractive system, calculating a 147 kΩ critical insulation resistance using the FSAE 250 Ω/V threshold to guide fault resistance selection and circuit design.

Morgan's Metronome v1.0 — July 2026 - September 2026

Arduino UNO, C++, KiCad, PCB Design, Embedded Systems

- Designed and programmed an embedded digital metronome in C++ using an Arduino UNO, with adjustable 40–180 BPM, potentiometer control, push button inputs, and a 16×2 LCD.
- Implemented non blocking timing with millis() to maintain beat intervals while keeping tempo and button inputs responsive.
- Prototyped and debugged the circuit on a breadboard before transferring it into an Arduino UNO compatible shield in KiCad.

Experience

Electrical and Firmware Team Member — September 2026 - Present

University of Waterloo Formula Electric — *Waterloo, Ontario*

- Collaborating with electrical and firmware engineering teammates in a multidisciplinary Formula SAE design team, contributing to circuit design and technical evaluations of vehicle electrical systems using Altium and LTSpice.
- Contributing to the redesign of the vehicle's Telematics Communication Unit by reviewing the existing cellular modem/SIM based architecture and investigating alternative wireless technologies for more reliable, lower cost real time telemetry and post run data analysis.

Badminton Program Instructor — August 2025 - May 2026

City of Richmond Hill - Rouge Woods Community Centre — *Richmond Hill, Ontario*

- Coached beginner and intermediate badminton classes of 15 students, tailoring lesson plans to individual student needs.

Education

University of Waterloo — September 2026 - April 2031

Bachelor of Applied Science, Honours Electrical Engineering (Co-op) — *Waterloo, Ontario*